

Let α, β be chosen (α = stepsize, β = momentum parameter)
 So that $R(\alpha, \beta) < 1$. Then for all $k \geq 0$,

$$\mathbb{E} \|y_{k+1} - x^*\| \leq R(\alpha, \beta)^k \|y_1 - x^*\| + \frac{\alpha \sqrt{(1+\beta)^2 + 1}}{1 - R(\alpha, \beta)} \sigma.$$

Where, $R(\alpha, \beta) = \max_{\lambda \in [0, 1]} R_\lambda(\alpha, \beta)$

$$R_\lambda(\alpha, \beta) = \frac{1}{\sqrt{2}} \left(C_\lambda(\alpha, \beta) + \sqrt{\tilde{\Delta}_\lambda(\alpha, \beta)} \right)^{1/2}$$

$$\tilde{\Delta}_\lambda(\alpha, \beta) = C_\lambda(\alpha, \beta)^2 - 4\beta^2(1 - \alpha\lambda)^2$$

$$C_\lambda(\alpha, \beta) = (1 - \alpha(1 + \beta)\lambda)^2 + \alpha^2\lambda^2 + \beta^2(\beta^2 + 1)$$

$$\sigma = \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x^*)\|.$$

This was theorem 4 in the paper.

We wanted this in the form of expectation of function values.

We have the following from the paper:

$$z_{k+1} = y_{k+1} - x^* \quad \left(\begin{array}{l} z_1 = x_0 - x^* \\ \vdots \end{array} \right)$$

$$v_k = x_k - x_{k-1} \quad \left(\begin{array}{l} v_0 = 0 \\ \vdots \end{array} \right)$$

$$y_{k+1} = x_k + \beta(x_k - x_{k-1}) \quad \left(\begin{array}{l} y_1 = x_0 \\ \vdots \end{array} \right)$$

$$\|z_k\| \leq \sum_{i=1}^k v_{k,i} \|\nabla f_i(x^*)\| \Rightarrow z_k = \sum_{i=1}^k v_{k,i} \nabla f_i(x^*); \mathbb{E}[z_k] = 0; \mathbb{E}[\|z_k\|] = \sigma.$$

It is given, assuming α, β such that $R(\alpha, \beta) < 1$:

$$\|A_k \dots A_{j+1}\| \leq \prod_{l=j+1}^k \|A_l\| \leq R(\alpha, \beta)^{k-j} \quad \forall k, j+1. \quad \text{--- (i)}$$

From L -smoothness, (descent lemma)

$$f(y_{k+1}) \leq f(x^*) + \langle \nabla f(x^*), y_{k+1} - x^* \rangle + \frac{L}{2} \|y_{k+1} - x^*\|^2.$$

↳ 0, making the entire term 0.

$$\Rightarrow f(y_{k+1}) - f(x^*) \leq \frac{L}{2} \|y_{k+1} - x^*\|^2.$$

Taking expectation:

$$\mathbb{E}[f(y_{k+1}) - f(x^*)] \leq \frac{L}{2} \mathbb{E}[\|y_{k+1} - x^*\|^2] \quad \text{--- (ii)}$$

From (31) in appendix,

$$\begin{bmatrix} \eta_{k+1} \\ v_k \end{bmatrix} = A_k \dots A_1 \begin{bmatrix} \eta_1 \\ v_0 \end{bmatrix} - \alpha \begin{bmatrix} (1+\beta)I \\ I \end{bmatrix} z_k - \alpha \sum_{j=1}^{k-1} (A_k \dots A_{j+1}) \begin{bmatrix} (1+\beta)I \\ I \end{bmatrix} z_j$$

where $A_k = \begin{bmatrix} I - \alpha(1+\beta)\tilde{H}_k & \beta^2 I \\ -\alpha\tilde{H}_k & \beta I \end{bmatrix}$, $\tilde{H}_k = \text{hessian}$ (30)

Taking squared norm & expectation & using (32)

$$\mathbb{E} \left[\left\| \begin{bmatrix} \eta_{k+1} \\ v_k \end{bmatrix} \right\|^2 \right] \leq \underbrace{R(\alpha, \beta)^{2k}}_{\text{from (i)}} \underbrace{\|x_0 - x^*\|}_{\text{euclidean dist.}} + \frac{\alpha^2 ((1+\beta)^2 + 1)^2 \sigma^2}{1 - R(\alpha, \beta)^2}$$

$$\Rightarrow \mathbb{E} [\|y_{k+1} - x^*\|^2] \leq R(\alpha, \beta)^{2k} \|x_0 - x^*\| + \frac{\alpha^2 ((1+\beta)^2 + 1)^2 \sigma^2}{1 - R(\alpha, \beta)^2} \quad \text{(iii)}$$

(expansion for reference)

(32) gives

$$\left\| \begin{bmatrix} \eta_{k+1} \\ v_k \end{bmatrix} \right\| \leq R(\alpha, \beta)^k \left\| \begin{bmatrix} \eta_1 \\ v_0 \end{bmatrix} \right\| + \alpha \sqrt{(1+\beta)^2 + 1} \sum_{j=1}^k R(\alpha, \beta)^{k-j} \|z_k\|$$

Squaring gives

$$\left\| \begin{bmatrix} \eta_{k+1} \\ v_k \end{bmatrix} \right\|^2 \leq \underbrace{R(\alpha, \beta)^{2k} \left\| \begin{bmatrix} \eta_1 \\ v_0 \end{bmatrix} \right\|^2}_A + \underbrace{\alpha^2 ((1+\beta)^2 + 1)^2 \sum_{j=1}^k R(\alpha, \beta)^{2(k-j)} \|z_k\|^2}_B + 2 \langle A, B \rangle$$

this cross-term vanishes when taking $\mathbb{E}$, since $\mathbb{E}[z_k] = 0$.

Taking expectation gives

$$\mathbb{E} [\|y_{k+1} - x^*\|^2] \leq A + B$$

$$\sum_{j=1}^k R(\alpha, \beta)^{2(k-j)} \leq \frac{1}{1 - R(\alpha, \beta)^2}$$

using geometric series

$$\sum_{j=1}^{k-1} R^{2j} = \frac{1 - R^{2k}}{1 - R^2} \leq \frac{1}{1 - R^2}$$

Finally, from (ii) & (iii),

$$\mathbb{E} [f(y_{k+1}) - f(x^*)] \leq \frac{L}{2} \left[R(\alpha, \beta)^{2k} \|x_0 - x^*\|^2 + \frac{\alpha^2 ((1+\beta)^2 + 1)^2 \sigma^2}{1 - R(\alpha, \beta)^2} \right] \#$$